C

SAAT / GAT QUANTITATIVE (ENGLISH) • UNIT 3.01

GAT Quantitative Comparisons & Fast Geometry

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

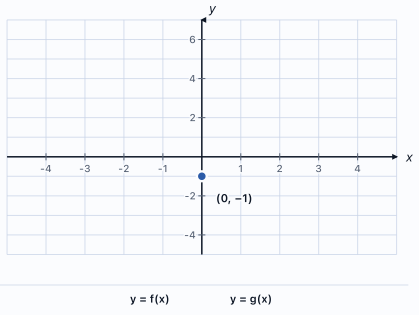
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 D	Q3 A	Q4 B	Q5 C
Q6 D	Q7 A	Q8 B	Q9 A	Q10 B
Q11 C	Q12 D	Q13 A	Q14 B	Q15 C
Q16 D	Q17 A	Q18 B	Q19 C	Q20 D
Q21 A	Q22 B	Q23 C	Q24 D	Q25 A

GAT Quantitative Comparisons & Fast Geometry Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 10x + 38$ in the domain of GAT Quantitative Comparisons & Fast Geometry. By completing the square $f_1(x) = (x - h)^2 + k$, find the sum $k + f_1(7)$.



A 35

B 17

C 30 ✓

D 28

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Complete the square: $x^2 - 10x + 38 = (x - 5)^2 - 25 + 38 = (x - 5)^2 + 13$, giving minimum value $k = 13$. **[M1]**
2. Evaluate at $x = 7$: $f_1(7) = (2)^2 + 13 = 17$. **[M1]**
3. Sum the two quantities: $k + f_1(7) = 13 + 17 = 30$. **[A1]**

- **Final Verified Answer:** 30.

2 MCQ

A mathematical model in GAT Quantitative Comparisons & Fast Geometry is defined by the cubic function $y_2(x) = 4x^3 - 5x^2 + 15x - 10$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

B 78

C 41

D 93 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 10x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 12(9) - 10(3) + 15 = 93$. **[M1][A1]**

- **Final Verified Answer:** 93.

3 MCQ

In GAT Quantitative Comparisons & Fast Geometry, two displacement vectors are given by $\mathbf{u}_3 = \langle 5, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.

A 78 ✓

B -11

C 89

D 190

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (5)(9) + (-8)(7) = 45 + (-56) = -11$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 5^2 + (-8)^2 = 89$. **[M1]**
3. Combine: $-11 + 89 = 78$. **[A1]**

- **Final Verified Answer:** 78.

4 MCQ

The linear growth profile shown in the diagram for GAT Quantitative Comparisons & Fast Geometry follows an arithmetic progression with first term $a_1 = 6$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 90

B 720 ✓

C 810

D 1350

WORKED SOLUTION & VERIFICATION

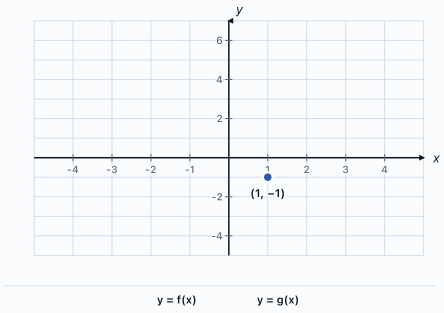
Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 6, d = 6$: $S_{15} = \frac{15}{2}[12 + 14 \times 6] = \frac{15}{2} \times 96 = 720$. **[M1][A1]**

- **Final Verified Answer:** 720.

5 MCQ

In GAT Quantitative Comparisons & Fast Geometry, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 20x)\,\mathrm{d}x$.



A 112

B 108

C 88 ✓

D 45

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Integrate term by term: $\int (18x^2 + 20x)\,\mathrm{d}x = [6x^3 + 10x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 10(2)^2) - 0 = 48 + 40 = 88$. **[M1][A1]**

- **Final Verified Answer:** 88.

6 MCQ

In GAT Quantitative Comparisons & Fast Geometry, consider the quadratic function $f_6(x) = x^2 - 12x + 121$. Express $f_6(x)$ in the completed-square form $(x - h)^2 + k$ and compute the exact value of $k + f_6(13)$.

A 225

B 134

C 212

D 219 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Complete the square: $x^2 - 12x + 121 = (x - 6)^2 - 36 + 121 = (x - 6)^2 + 85$, giving minimum value $k = 85$. **[M1]**
2. Evaluate at $x = 13$: $f_6(13) = (7)^2 + 85 = 134$. **[M1]**
3. Sum the two quantities: $k + f_6(13) = 85 + 134 = 219$. **[A1]**

- **Final Verified Answer:** 219.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 6x^2 + 33x - 17$ studied in GAT Quantitative Comparisons & Fast Geometry. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

- **A** 705 ✓
- **B** 801
- **C** 672
- **D** 117

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 12x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 12(64) - 12(8) + 33 = 705$. **[M1][A1]**

- **Final Verified Answer:** 705.

8 MCQ

In GAT Quantitative Comparisons & Fast Geometry, two displacement vectors are given by $\mathbf{u}_8 = \langle 6, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

- **A** 13
- **B** 98 ✓
- **C** 85
- **D** 168

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (GAT Quantitative Comparisons & Fast Geometry):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (6)(8) + (-7)(5) = 48 + (-35) = 13$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 6^2 + (-7)^2 = 85$. **[M1]**
3. Combine: $13 + 85 = 98$. **[A1]**

- **Final Verified Answer:** 98.

9 MCQ

Compare the two quantities when $x = \frac{1}{3}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

- **A** **Quantity A is greater** ✓
- **B** Quantity B is greater
- **C** The two quantities are equal
- **D** The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

For $0 < x < 1$, taking square roots increases the number while squaring decreases it: $\sqrt{1/3} > (1/3)^2$. Hence Quantity A is greater.

10 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{4^2 + 7^2}$ | Quantity B: $4 + 7$

- A Quantity A is greater
- **B Quantity B is greater** ✓
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

Since $4 > 0$ and $7 > 0$, $(4 + 7)^2 = 4^2 + 7^2 + 2(4)(7) > 4^2 + 7^2$. Taking positive square roots shows $\sqrt{4^2 + 7^2} < 4 + 7$, so Quantity B is greater.

11 MCQ

Compare the two quantities: Quantity A: 12% of 35 | Quantity B: 35% of 12

- A Quantity A is greater
- B Quantity B is greater
- **C The two quantities are equal** ✓
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

By percentage commutativity, $\frac{12}{100} \times 35 = \frac{420}{100} = \frac{35}{100} \times 12$. Both quantities are identically equal.

12 MCQ

Given $x^2 = 16$ and $y = 3$, compare: Quantity A: x | Quantity B: y

- A Quantity A is greater
- B Quantity B is greater
- C The two quantities are equal
- **D The relationship cannot be determined from the information given** ✓

WORKED SOLUTION & VERIFICATION

From $x^2 = 16$, x can be either $+4$ (making $x > y$) or -4 (making $x < y$). Because two valid cases yield opposite orderings, the relationship cannot be determined.

13 MCQ

Compare the two quantities when $x = \frac{1}{4}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

- **A Quantity A is greater** ✓
- B Quantity B is greater
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

For $0 < x < 1$, taking square roots increases the number while squaring decreases it: $\sqrt{1/4} > (1/4)^2$. Hence Quantity A is greater.

14 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{5^2 + 8^2}$ | Quantity B: $5 + 8$

- A Quantity A is greater
- **B Quantity B is greater** ✓
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

Since $5 > 0$ and $8 > 0$, $(5 + 8)^2 = 5^2 + 8^2 + 2(5)(8) > 5^2 + 8^2$. Taking positive square roots shows $\sqrt{5^2 + 8^2} < 5 + 8$, so Quantity B is greater.

15 MCQ

Compare the two quantities: Quantity A: 14% of 38 | Quantity B: 38% of 14

- A Quantity A is greater
- B Quantity B is greater
- **C The two quantities are equal** ✓
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

By percentage commutativity, $\frac{14}{100} \times 38 = \frac{532}{100} = \frac{38}{100} \times 14$. Both quantities are identically equal.

16 MCQ

Given $x^2 = 25$ and $y = 4$, compare: Quantity A: x | Quantity B: y

- A Quantity A is greater
- B Quantity B is greater
- C The two quantities are equal
- **D The relationship cannot be determined from the information given** ✓

WORKED SOLUTION & VERIFICATION

From $x^2 = 25$, x can be either $+5$ (making $x > y$) or -5 (making $x < y$). Because two valid cases yield opposite orderings, the relationship cannot be determined.

17 MCQ

Compare the two quantities when $x = \frac{1}{5}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

- **A Quantity A is greater** ✓
- B Quantity B is greater
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

For $0 < x < 1$, taking square roots increases the number while squaring decreases it: $\sqrt{1/5} > (1/5)^2$. Hence Quantity A is greater.

18 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{6^2+9^2}$ | Quantity B: $6+9$

- A Quantity A is greater
- **B Quantity B is greater** ✓
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

Since $6>0$ and $9>0$, $(6+9)^2=6^2+9^2+2(6)(9)>6^2+9^2$. Taking positive square roots shows $\sqrt{6^2+9^2}<6+9$, so Quantity B is greater.

19 MCQ

Compare the two quantities: Quantity A: 16% of 41 | Quantity B: 41% of 16

- A Quantity A is greater
- B Quantity B is greater
- **C The two quantities are equal** ✓
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

By percentage commutativity, $\frac{16}{100}\times 41=\frac{656}{100}=\frac{41}{100}\times 16$. Both quantities are identically equal.

20 MCQ

Given $x^2=36$ and $y=5$, compare: Quantity A: x | Quantity B: y

- A Quantity A is greater
- B Quantity B is greater
- C The two quantities are equal
- **D The relationship cannot be determined from the information given** ✓

WORKED SOLUTION & VERIFICATION

From $x^2=36$, x can be either $+6$ (making $x>y$) or -6 (making $x<y$). Because two valid cases yield opposite orderings, the relationship cannot be determined.

21 MCQ

Compare the two quantities when $x=\frac{1}{6}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

- **A Quantity A is greater** ✓
- B Quantity B is greater
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

For $0<x<1$, taking square roots increases the number while squaring decreases it: $\sqrt{1/6}>(1/6)^2$. Hence Quantity A is greater.

22 MCQ

Compare the two positive expressions: Quantity A: $\sqrt{7^2 + 10^2}$ | Quantity B: $7 + 10$

- A Quantity A is greater
- **B Quantity B is greater** ✓
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

Since $7 > 0$ and $10 > 0$, $(7 + 10)^2 = 7^2 + 10^2 + 2(7)(10) > 7^2 + 10^2$. Taking positive square roots shows $\sqrt{7^2 + 10^2} < 7 + 10$, so Quantity B is greater.

23 MCQ

Compare the two quantities: Quantity A: 18% of 44 | Quantity B: 44% of 18

- A Quantity A is greater
- B Quantity B is greater
- **C The two quantities are equal** ✓
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

By percentage commutativity, $\frac{18}{100} \times 44 = \frac{792}{100} = \frac{44}{100} \times 18$. Both quantities are identically equal.

24 MCQ

Given $x^2 = 49$ and $y = 6$, compare: Quantity A: x | Quantity B: y

- A Quantity A is greater
- B Quantity B is greater
- C The two quantities are equal
- **D The relationship cannot be determined from the information given** ✓

WORKED SOLUTION & VERIFICATION

From $x^2 = 49$, x can be either $+7$ (making $x > y$) or -7 (making $x < y$). Because two valid cases yield opposite orderings, the relationship cannot be determined.

25 MCQ

Compare the two quantities when $x = \frac{1}{7}$: Quantity A: $\sqrt{x}$ | Quantity B: x^2

- **A Quantity A is greater** ✓
- B Quantity B is greater
- C The two quantities are equal
- D The relationship cannot be determined from the information given

WORKED SOLUTION & VERIFICATION

For $0 < x < 1$, taking square roots increases the number while squaring decreases it: $\sqrt{1/7} > (1/7)^2$. Hence Quantity A is greater.